Scalable Extraction of Author-Stated Research Gaps: Towards Automated Idea Generation

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Abstract

Research gaps stated by authors could feed automated idea generation, but they are rare and scattered in papers. Asking a large language model (LLM) to judge every sentence, or a whole paper at once, is expensive, and long inputs are hard for smaller models. We propose a hierarchical three-stage funnel that splits the task: rules drop unlikely text, a zero-shot natural-language-inference classifier flags candidate gaps, and an LLM filters the candidates as source sentences. A single instruction sets the filter to favour precision, balance or recall. We evaluated the funnel on two human-labelled benchmarks, SAL (limitations in randomized-trial reports) and FWS (future work in natural language processing papers). Because each labels only one gap type in one field, we also built a corpus of 230 arXiv and biomedical papers in which two LLMs labelled limitations, future work and research gaps, with a third LLM pass resolving disagreements, and evaluated on it as well. In its high-precision setting, the LLM filter more than doubled precision on SAL, from 29.1% after the classifier to 66.2%, while keeping 84.9% recall, and on every dataset the three settings traded precision for recall as intended. The funnel used 3.7–13 times fewer LLM input tokens than judging every sentence.

Keywords: research gap extraction; limitations; future work; scientific text mining; large language models; natural language inference; zero-shot classification; LLM filtering

1 Introduction

Research papers often tell readers where a study falls short and what should be investigated next. In clinical research, self-acknowledged limitations help readers interpret findings in light of study design, data and applicability [1]. In other fields, future-work statements name extensions, open questions and further evaluations that authors consider worth pursuing [2, 3]. Authors also point to unsolved problems in their field when they motivate a study. We call these statements *author-stated research gaps*: limitations of the authors' own work, future-work directions they propose, and open problems they identify. Collected across many papers, such statements are a candidate input for automated idea generation, because each one is a claim, made by a domain expert, about what is still missing.

Extracting these statements at scale is harder than it looks. Gap statements are sparse: in the corpora studied here they make up 2.6% of the sentences of full RCT reports, 5.7% of full arXiv and biomedical papers, and 14.6% of the sentences of conclusion sections. They are also scattered. Authors may put limitations in a dedicated section, inside a discussion, or next to a methodological choice, and in computer-science papers many gaps sit in the methods and

experiments sections rather than at the end. A sentence that criticises prior work, restates a result or describes a mitigation can look like a gap without being one. Finally, the output must stay attributable: a system that generates plausible criticism is answering a different question from one that finds what the authors actually wrote [4, 5].

Large language models (LLMs) can judge whether a sentence is a gap, but sending every sentence of a paper to an LLM is costly, and sending a whole paper leaves the model to find a few gap sentences in a very long input. Earlier extraction work suggests a cheaper division of labour. Clinical limitation recognition uses headings, lexical cues and supervised classifiers [1]; later work on randomized controlled trial (RCT) reports separates limitation detection from limitation typing [6]; and future-work recognition separates locating sentences from classifying them [2, 3]. These studies motivate placing cheap structural and lexical evidence before more flexible semantic judgement.

We study a three-stage funnel built on this idea. Stage A discards text that is unlikely to contain gaps, Stage B flags candidate sentences with a classifier, and Stage C asks an LLM to filter the candidates. Every stage selects existing sentences and never rewrites them, so each output can be traced to its source. Two design choices are central to this study. First, Stage B is a zero-shot natural-language-inference (NLI) cross-encoder that needs no task-specific training labels, so the same funnel runs on biomedical and computer-science papers. Second, Stage C has three modes set by a single instruction: high precision, balanced and high recall. Because the extracted gaps are meant to feed idea generation, where a false gap wastes effort, we treat precision as the primary goal and use high precision as the main mode.

We evaluate the funnel on two benchmarks with human labels, randomized-trial reports with limitation labels (SAL) [6] and future-work sentences from natural language processing (NLP) papers (FWS) [3], and on a corpus of 230 arXiv and biomedical papers that we labelled for three gap types with an ensemble of LLMs. Results on the LLM-labelled corpus are reported as provisional until human validation, which is in progress, is complete.

Research questions.

- **RQ1 (Stage contribution).** How much does each stage change precision and recall on full-paper sentence streams?
- **RQ2 (Precision control).** Can a single instruction set the precision–recall trade-off of the LLM filter, and do the settings behave consistently across datasets?
- **RQ3 (Classifier and cost).** Does a training-free NLI classifier suffice for Stage B compared with supervised classifiers, and how much LLM input does the funnel save?

Contributions.

1. A source-preserving funnel for author-stated research gaps with a training-free Stage B and an LLM filter whose precision–recall trade-off is set by one instruction (Section 3).
2. An evaluation on two human-labelled benchmarks, including a check on SAL development papers on which no prompt was written (they were used only to set the classifier threshold), and on a 230-paper, three-type LLM-labelled corpus, with paper-level confidence intervals and paired comparisons (Section 4).
3. Evidence that the filter modes behave as intended on every dataset: high precision mode reaches 66.2% precision on SAL, 76.5% on FWS and 86.6% against the LLM labels, and the training-free classifier comes within 0.021 F_1 of a supervised PubMedBERT on SAL, a difference that is not statistically significant, although the supervised classifier is 8.4 points more precise.

4. Two findings about evaluation data: letter case alone reveals the label in the public FWS corpus, and SAL and our LLM labels disagree on most sentences the filter keeps that SAL marks negative (Sections 4.7 and 4.6).

The scope is extraction. Whether the extracted gaps are still open, and whether they lead to useful research ideas, are separate questions for future work.

Section 2 reviews related work, Section 3 describes data, funnel and evaluation, Section 4 reports results, Section 5 discusses them and their limitations, and Section 6 concludes.

2 Related Work

Gap2Idea builds on research into scientific discourse classification, the recognition of author-stated limitations and future work, and LLM-based analysis of scholarly documents. Across these areas, similar terminology can refer to different tasks: locating an existing statement, inferring a missing piece of knowledge, or generating a new criticism or research direction. We organise the discussion around these distinctions and explain how each line of work informs our extraction pipeline. Table 1 compares the closest studies by target, method and input unit.

2.1 Scientific discourse and sentence classification

Scientific discourse annotation provides a foundation for identifying statements by their role within a paper. Argumentative zoning represents the rhetorical function of sentences and their relationship to the authors' argument [7]. Liakata et al. [8] distinguish this rhetorical perspective from Core Scientific Concepts (CoreSC), which represent components of an investigation, including its goals, methods, observations and conclusions. The Dr. Inventor corpus brings the discussion closer to our targets: its 40 computer-graphics articles have a Future Work category in the argumentative-structure layer and a Limitation category among the crosswise features [9]. These annotations establish a full-text precedent for representing both kinds of statement, although they do not constitute a single research-gap detection task.

Abstract sentence classification supplies further resources and modelling approaches. PubMed 200k RCT and CSAbstruct support the recognition of abstract roles in biomedical and computer-science publications, respectively [10, 11]. MAZEA allows multiple rhetorical labels per sentence [12], while Emerald 110k extends coverage across disciplines and includes a combined Research limitations/implications heading [13]. Such resources provide useful discourse supervision, but their labels and document scope differ from ours: a conclusion sentence need not state a gap, and a combined limitations/implications label also covers other content.

More targeted rhetorical schemes make the overlap explicit. Sci-Sentence includes Research Gap and Limitation among seven categories intended to support literature-review analysis [14]. Its benchmark samples sentences from introductions, related-work sections and limitations sections. Its Research Gap category overlaps with our open-problem target, whereas its Limitation definition concerns weaknesses of a previous study rather than exclusively the authors' own work. This difference in attribution matters when deciding which statements count as positives.

Context is another shared concern. Hierarchical sentence classifiers model surrounding discourse [15], and Brack et al. [16] investigate cross-domain sequential classification through multi-task learning. Lan et al. [17] extend sequential classification to LLMs and multi-label outputs, while Zhang and Zhang [18] use context-enhanced representations to identify problem and method sentences. These studies motivate considering neighbouring text when a sentence is ambiguous. In Gap2Idea, local context is supplied to the final LLM filter, which decides whether a previously selected candidate expresses the target statement.

2.2 Recognition of limitations and future work

Work on self-acknowledged limitations directly addresses one of our three targets. Kilicoglu et al. [1] formulate limitation recognition in clinical research as sentence-level binary classification and investigate both rules and learned classifiers. Their use of section headings and limitation zones provides a direct precedent for structural localisation. Lan et al. [6] subsequently annotate limitations in 200 randomized controlled trial (RCT) publications and separate detecting a limitation from assigning its type. Their fine-tuned PubMedBERT detector achieves a reported F_1 of 0.821 under their evaluation protocol. SAL supplies our human-labelled limitation benchmark, and the PubMedBERT model family supplies a supervised comparison. We distinguish that published score from our full-paper evaluation because the evaluated sentence pool and class balance affect precision and F_1.

Future-work recognition has a parallel history. Zhu et al. [2] use BERT-based recognition to identify future-research sentences in JASIST articles and cluster the extracted statements. The ACL FWS-RC corpus provides annotations and a taxonomy for future-work sentences in NLP papers [19]. Using this resource, Zhang et al. [3] compare lexical and neural approaches, reporting their strongest binary recognition result with Bernoulli naive Bayes (macro-F_1 of 0.9073). Their recognition and direction-classification tasks are distinct, as are macro-F_1 and the positive-class F_1 used in our main evaluation. These published scores therefore provide context rather than a ranking against our paper-disjoint experiments.

Extracted statements also support analysis beyond detection. Suray et al. [20] examine how subsequent usable-security research engages with previously stated future work. More recently, Thaepprasit et al. [21] analyse self-reported limitations in ACL and EMNLP papers through iterative human–AI qualitative coding. These studies show how author-written statements can support research-trend and reporting analyses. Gap2Idea addresses the preceding extraction step; it does not establish whether a stated gap remains unresolved or whether a proposed direction has subsequently been pursued.

2.3 LLM extraction, inferred gaps and generated critique

LLMs are increasingly used both to extract statements and to produce new assessments of research. BAGELS constructs limitation data from ACL, NeurIPS and PeerJ papers, supplements author-written material with external reviews, and studies retrieval-augmented limitation generation [22]. FutureGen similarly uses extracted future-work material to support the generation and evaluation of new directions [23]. Cheng et al. [24] directly investigate prompted LLM extraction of future-work sentences in NLP papers and connect the extracted material to keyword-based research-direction analysis. Their pipeline selects relevant sections before LLM processing. Together, these studies establish that both LLM extraction and structural selection already have close precedents.

Knowledge-gap identification also includes an inference task. GAPMAP explicitly distinguishes stated biomedical knowledge gaps from missing knowledge inferred from context, evaluating LLMs in paragraph-level and full-paper settings [25]. This distinction is central to our scope: we select an existing sentence that articulates a gap and preserve its wording. We do not ask the model to construct an implicit gap that has no corresponding author-written statement.

Limitation generation and scientific critique involve a further change in the source of the output. LimGen studies suggestive limitation generation using 4068 ACL papers and their associated author-written limitations [26]. Despite its similar name, LimitGen is a separate benchmark for evaluating LLM-generated limitations and feedback on AI research papers [4]. Tree-of-Concerns focuses specifically on unstated limitations through structured multi-agent critique [5]. These tasks can produce useful criticisms beyond those acknowledged by the authors. Our LLM instead returns a keep-or-drop decision for each candidate sentence; the resulting extraction remains attributable to the original text.

2.4 Position of this work

Our Stage B builds on established methodology. Yin et al. [27] formulate zero-shot text classification as textual entailment, allowing a label to be expressed as a natural-language hypothesis. We apply this formulation to gap types using an existing NLI model. Here, zero-shot means that the classifier is not fitted on gap-detection training examples; the model has already been trained on NLI data, and our decision threshold is selected with labelled development data.

Our contribution is the evaluation of a particular combination: structural localisation, NLI candidate selection and a context-aware LLM filter with three precision–recall instructions. We examine how these stages interact, how the filter settings behave across datasets, and how candidate selection changes LLM input cost. The evaluation covers both full papers and the section-level FWS resource, keeping their processing units and label sources explicit. Human-labelled limitation and future-work benchmarks support those two targets; findings for the joint three-type task remain provisional because that corpus is LLM-labelled. The resulting evidence concerns source-preserving extraction; assessing the quality of research ideas derived from these statements remains a separate task.

3 Materials and Methods

Figure 1 gives an overview of the method. This section first defines the task and the data, then describes the three stages of the funnel, explains how their settings were chosen, and introduces the comparison systems and the evaluation protocol.

3.1 Task

Given all sentences of a paper, the system returns the sentences in which the authors state a research gap. We distinguish three types of gap:

- **Limitation**: a weakness, caveat, bias, constraint or assumption the authors acknowledge about their own work;
- **Future work**: an explicit direction the authors propose for future research;
- **Research gap**: an unsolved problem, missing capability or open challenge the authors point to in the field or in prior work.

The system never rewrites or summarises: each output is an unmodified source sentence. Evaluation is binary at the sentence level. A returned sentence counts as correct if the reference labels mark it as a gap of any type that the benchmark covers; the predicted type is not scored. SAL labels only limitations and FWS only future work, so on these two benchmarks the funnel is asked for that single type (Section 3.3).

3.2 Data

We use three datasets (Table 2): two public benchmarks with human labels, each covering one gap type, and a corpus of 230 papers that we labelled for all three types with LLMs. All splits are by paper, so the sentences of one paper never appear in two splits.

SAL. SAL contains 200 RCT reports in which annotators marked the sentences that state a limitation; it is split into 120 training, 40 development and 40 test papers [6, 28]. We call the test split SAL-40 and the development split SAL dev. We score every sentence of each paper, not a section-filtered subset. To apply section rules in Stage A, we recovered the section type of each sentence by matching it to the PubMed Central BioC version of its article. Sentences of fewer

Table 1: Selected related work, ordered by year of the cited version. Target categories describe the relevant part of each study; other rhetorical labels are omitted where applicable. When multiple target categories apply, they are listed in the order Limitations, Future work and Research gaps. "Input unit" is the input each method processes.

Study	Domain	Target categories	Approach	Input unit	Relation to this study
Dr. Inventor, 2016 [9]	Computer graphics	Limitations; Future work	Human multi-layer annotation	Sentences, full articles	Full-text precedent with both statement types
Kilicoglu et al., 2018 [1]	Clinical research	Limitations	Rules; supervised classifiers	Sentences	Precedent for section-based localisation
Zhu et al., 2019 [2]	Information science	Future work	BERT classification; clustering	Sentences, full articles	Links recognition to thematic analysis
ACL FWS-RC, 2020 [19]	NLP	Future work	Annotation; taxonomy	Sentences	Source of FWS labels
Zhang et al., 2022 [3]	NLP	Future work	Lexical and neural recognition	Sentences	FWS recognition benchmark
LimGen, 2024 [26]	NLP	Limitations	LLM generation	Paper	Author limitation sections as references
Lan et al. (SAL), 2024 [6]	RCT reports	Limitations	PubMedBERT detection and typing	Sentences	Source of SAL; supervised comparison
Lan et al. (multi-label), 2024 [17]	Several sciences	Rhetorical roles	LLM prompting and fine-tuning	Abstract sequence	Context-aware LLM labelling
Zhang & Zhang, 2024 [18]	NLP	Problems; Methods	Context-enhanced transformer	Sentence with context	Local context for sparse targets
Brack et al., 2024 [16]	Several sciences	Rhetorical roles	Cross-domain multi-task learning	Sentence sequence	Transfer across domains
Suray et al., 2024 [20]	Usable security	Future work	Statement and citation analysis	Statement	Downstream use of extracted directions
BAGELS, 2025 [22]	NLP; ML; biology	Limitations	Extraction; RAG generation	Paper; reviews	Extraction and generation benchmark
FutureGen, 2025 [23]	NLP; ML	Future work	LLM extraction; RAG generation	Paper	Extraction feeding generation
GAPMAP, 2025 [25]	Biomedicine	Knowledge gaps	LLM detection and inference	Paragraph to paper	Explicit versus inferred gaps
LimitGen, 2025 [4]	AI	Limitations	LLM critique; retrieval	Paper	Evaluates generated limitations
Sci-Sentence, 2025 [14]	Several sciences	Limitations; Research gaps	Encoder and LLM classifiers	Sentence	Gap labels; different attribution
Tree-of-Concerns, 2026 [5]	ML; NLP; vision	Limitations	Multi-agent critique	Paper	Unstated critique; different target
Cheng et al., 2026 [24]	NLP	Future work	Section selection; LLM extraction	Selected sections	LLM extraction with structural selection
Thaepprasit et al., 2026 [21]	NLP	Limitations	Human–AI qualitative coding	Limitation sections	Downstream analysis of reported limitations
This study	RCT reports; NLP; arXiv	Limitations; Future work; Research gaps	Rules, zero-shot NLI, LLM filter with three modes	Sentence with context	Measures stage contribution, precision control and cost

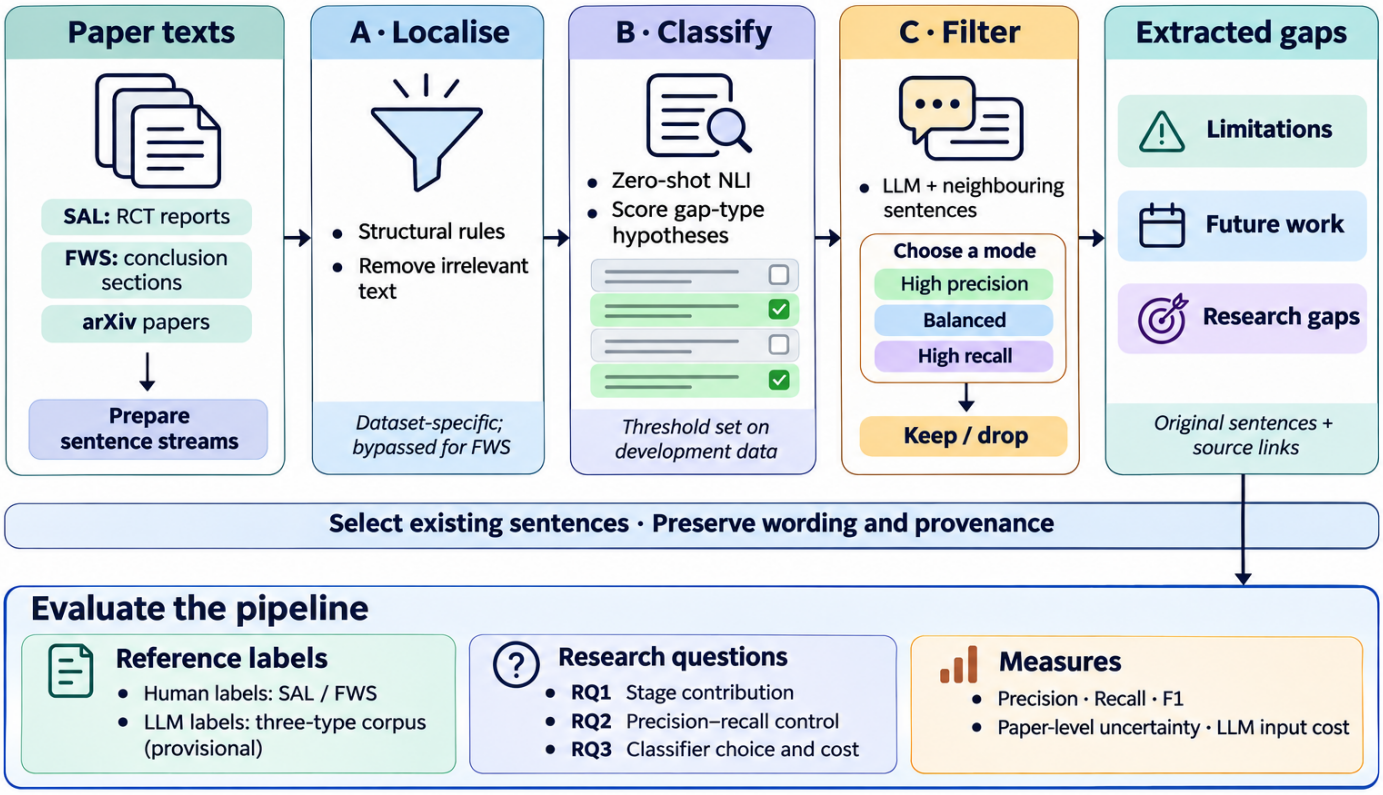


Figure 1: Overview of Gap2Idea. Each paper is split into sentences. Stage A removes text that is unlikely to contain gaps (its rules depend on the dataset; FWS skips it), Stage B selects candidate sentences with a zero-shot NLI classifier, and Stage C asks an LLM to keep or drop each candidate. The three filter modes are alternative instructions for Stage C, not consecutive steps. Every output is an original sentence with its source. The evaluation (bottom) uses human labels on SAL (limitations) and FWS (future work), and provisional LLM labels for all three gap types.

Table 2: Evaluation data. Positives are sentences labelled with a gap type the benchmark covers.

Dataset (split)	Labels	Gap types	Papers	Sentences	Positives	Use
SAL (test)	Human	Limitation	40	7968	205	Main evaluation
SAL (dev)	Human	Limitation	40	7729	180	Threshold; held-out check
FWS (test)	Human	Future work	209	1506	220	Evaluation
FWS (dev)	Human	Future work	418	3005	418	Threshold
Gold corpus (test)	LLM ensemble	All three	34	11,741	664	Evaluation
Gold corpus (validation)	LLM ensemble	All three	34	11,201	699	Design choices; thresholds

than three words were removed; none of the removed test sentences is a limitation. The training split is used only to train the supervised classifiers we compare against (Section 3.5).

FWS. The FWS recognition corpus contains sentences from the conclusion sections of ACL papers, each labelled as future work or not [3, 19, 29]. Because it holds conclusion sections only, it has no full-paper structure; we use it to test the funnel on a second domain and a second gap type. We first removed every sentence that also occurs in the training data of an earlier classifier of ours. We then shuffled the papers once and took them in that order: the first 209 papers (1506 sentences) form the test sample, and the next 418 papers form the development sample. All FWS text is lowercased before any processing, because in this corpus letter case reveals the label (Section 4.7).

Gold corpus. To evaluate all three gap types together, we built a corpus of 230 papers and 68,719 sentences. It contains 190 arXiv computer-science and machine-learning papers, converted from PDF with PyMuPDF and split into sentences after removing reference lists, and the 40 SAL-40 papers with their original sentence boundaries. Its labels were produced in two steps:

1. *Independent labelling.* Two LLMs, Claude Opus 5.5 and Claude Sonnet 5.5, each labelled every sentence of every paper, in one call per paper, following a written guideline with the three definitions above. SAL labels were not shown to them. The 2960 sentences that both models marked with the same type form the high-confidence tier.

2. *Adjudication.* The remaining 1737 candidates, marked by only one model or with different types, were judged again by Claude Opus 5.5 with three sentences of context on each side. It confirmed 1502 of them.

The final corpus has 4462 gap sentences: 2234 research gaps, 1543 limitations and 685 future-work statements. The two labelling models agreed on whether a sentence is a gap with $\kappa = 0.778$ [30]; this measures agreement between models, not agreement with people. On the 40 SAL papers, the ensemble marked all but nine of SAL's 205 human-labelled limitations as a gap. Labelling cost about US$36. We call this corpus the *gold corpus* because its labels are our reference for three-type evaluation; since no person has checked them yet, all results on it are provisional. Papers were split into 162 training, 34 validation and 34 test papers. The test split has 664 gaps (220 limitations, 97 future work and 347 research gaps); 28 of its papers are from arXiv (527 gaps) and 6 are SAL papers (137 gaps).

3.3 The funnel

The funnel applies three stages in sequence, each keeping a subset of the sentences it receives. Figure 2 shows how many sentences each stage keeps on the gold test split; Section 4.1 analyses these numbers.

Stage A: localisation. Stage A removes text that is unlikely to contain gaps before any model is run. Its rules depend on the dataset, because papers differ in how reliably their sections show where gaps are stated.

- **SAL:** only sentences in Discussion and Conclusion sections are kept. This removes 76.3% of the sentences but keeps 203 of the 205 test limitations.

- **Gold corpus:** no section rule is applied, because in arXiv papers many gaps are stated in the methods and experiments sections (Section 5.3). Stage A only removes text that cannot be a gap statement, and we call it the *junk filter*. It drops titles, tables, figure captions and abbreviation lists; competing-interest, author-contribution, funding and

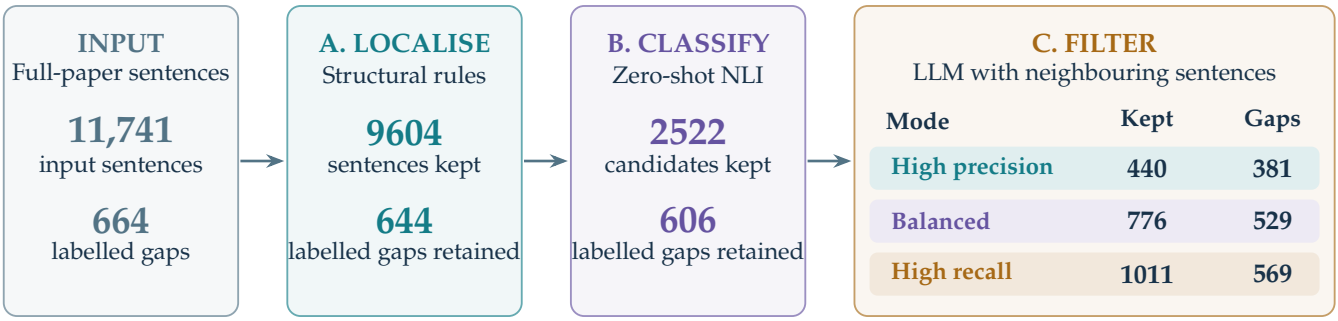


Figure 2: Sentence selection on the gold test split (34 papers, provisional LLM labels). The first three cards show the input and the sentences retained after Stages A and B. Stage C applies each alternative filter mode to the same candidates; "Kept" counts all selected sentences and "Gaps" counts selected sentences labelled as gaps. Each stage preserves the original text.

supplementary sections, where section types are known; bibliography-like strings; and malformed sentences (under 25 characters or six tokens, starting mid-word, containing dot leaders, or consisting mostly of non-letters).

- **FWS:** there is no Stage A. The corpus already consists of conclusion sections, and the junk filter, which rejects sentences that start in lower case, would remove almost all FWS text once it is lowercased.

Stage B: zero-shot NLI classifier. Stage B selects candidate sentences with a classifier that needs no gap-labelled training data. We use an NLI cross-encoder, ModernBERT-large fine-tuned on NLI data [31, 32], as a zero-shot classifier [27]: the model reads a sentence (the premise) together with a short statement of the label (the hypothesis) and estimates whether the sentence entails the statement. There is one hypothesis per gap type: "This is a limitation.", "This is future work." and "This is a research gap." A sentence's score for a hypothesis is the entailment probability divided by the sum of the entailment and contradiction probabilities. On the gold corpus, a sentence becomes a candidate if its highest score over the three hypotheses reaches a threshold τ, and its predicted type is the hypothesis with that highest score. On SAL only the limitation hypothesis is used, and on FWS only the future-work hypothesis. Inputs are truncated to 256 tokens. Section 3.4 describes how τ is set.

Stage C: LLM filter. Stage C asks an LLM to keep or drop each candidate. We use Qwen3.8-27B, self-hosted with vLLM [33]; this model was not used to create any labels. Candidates are sent in reading order, 20 per request. For each candidate, the model receives its predicted type, the sentence (up to 400 characters) and, as context, the previous and next sentences of the same paper (up to 600 characters). It returns a keep or drop verdict for every candidate as JSON constrained by a schema. Decoding uses temperature 0, seed 0 and no reasoning mode.

The system prompt (Appendix A) defines the three gap types, lists the kinds of sentences to drop (results, method descriptions, contribution claims, background, citations, captions and similar), and ends with one instruction that sets the mode. Only this final instruction differs between the modes:

- **High recall**: when genuinely uncertain, keep the sentence;
- **Balanced**: favour neither keeping nor dropping;
- **High precision**: keep a sentence only if it explicitly states the gap itself; drop general background, motivation, observed outcomes and passing remarks; and when uncertain, drop it.

For SAL and FWS, the definitions are narrowed to the benchmark's single type, and the other two types are added to the list of sentences to drop. The mode instructions are identical on all datasets.

3.4 How settings were chosen

Several settings were tuned. To make clear which data could have influenced each choice, we list where it was made.

- **NLI hypotheses.** We compared hypothesis variants (long definitions, short type names, a single combined hypothesis, adding the neighbouring sentences to the premise, and subtracting competing "not a gap" hypotheses) on one half of the gold validation papers and confirmed the choice on the other half. Short type names worked best; adding context or competing hypotheses made the results worse.
- **Thresholds.** τ is set on development data only. It is the highest value at which recall, counting the positives already removed by Stage A, is still at least 90%. This gives $\tau = 0.354$ on the gold validation split, $\tau = 0.7286$ on SAL dev and $\tau = 0.1922$ on FWS dev. Fixing recall in this way lets Stage B keep most gaps and leaves the removal of false positives to Stage C. The stored values are rounded; at the rounded SAL value one development limitation falls below τ, so SAL dev recall before Stage C is 89.4% instead of 90.0%.
- **Stage C model and prompt.** The LLM, the batch size, the use of context and the high-recall instruction were chosen during development on 196 candidates from gold test papers. We therefore also report gold results without these 196 sentences (Section 4.2). The balanced and high-precision instructions were written after reading the false positives of a high-recall run on gold test papers, and were then checked on the gold validation split before they were run on any test set. All three modes were run on every test set and all are reported, so no mode was selected by its test score.
- **Held-out check.** SAL dev was used only to set τ; no prompt was written or chosen on it. Running the full funnel on SAL dev therefore tests it on papers that did not influence any prompt.

3.5 Comparison systems

Supervised classifiers on SAL. To test whether a training-free Stage B is sufficient, we compare it with classifiers trained on SAL's own labels. We trained eight classifiers on the SAL training split. Three are fine-tuned encoders: PubMedBERT [34] (`BiomedNLP-BiomedBERT-base-uncased-abstract-fulltext`), SciBERT [35] and DistilBERT [36], each trained for three epochs (learning rate 3×10^{-5}, batch 32, class-weighted loss) with a decision threshold chosen on SAL dev. The other five are frozen sentence encoders [37, 38] with a logistic-regression classifier. PubMedBERT was the strongest of the eight and is used as the supervised Stage B; it is also the model family used by Lan et al. [6].

Other Stage B options on the gold corpus. On the gold corpus, we compare the NLI classifier with two alternatives: (i) a frozen bge-small encoder [38] with a logistic-regression classifier trained on the gold training split, and (ii) the same ModernBERT NLI model with a single long hypothesis that covers all gap types, used on its own without Stage C (threshold 0.75, chosen on validation).

Not yet included. Comparisons with whole-paper and every-sentence LLM prompting on the same test papers, and reproductions of published methods, are in progress and are not reported in this version.

3.6 Evaluation

Metrics. We report precision, recall and F_1 for the positive class at the sentence level. Recall is computed over all positive sentences in the test papers, including those removed by Stage A, so the losses of every stage count. Sentences from the same paper are not independent, so we resample papers rather than sentences: confidence intervals (CIs) are 95% percentile intervals from 2000 bootstrap resamples of papers [39]. To compare two systems, we score both on the same resampled papers and report the CI of their difference (a paired comparison); we call a difference significant when this interval excludes zero.

LLM input cost. We estimate the number of LLM input tokens with the `cl100k_base` tokenizer [40] on the prompts and items exactly as sent. Output tokens and the computation of Stages A and B are not counted. We compare the funnel with two alternatives that skip Stages A and B: one whole-paper call (the high-precision prompt followed by the numbered sentences of the paper) and judging every sentence in batches of 20 in the funnel's format.

Run-to-run stability. To measure how much the LLM verdicts vary between identical runs, we repeated the high-recall run on the gold test candidates.

3.7 Software and use of generative AI

Experiments used Python with PyTorch, Hugging Face Transformers, sentence-transformers, scikit-learn and vLLM; exact versions are pinned in the repository's lock file. The code is available at the repository listed in the Data Availability Statement. Generative AI is part of the method in two places: Claude Opus 5.5 and Claude Sonnet 5.5 produced the gold-corpus labels, and Qwen3.8-27B is the Stage C filter. The use of AI assistance in preparing the study and the manuscript is declared in the Acknowledgments.

4 Results

We report the results in the order of the research questions: what each stage contributes (RQ1, Section 4.1), how the three filter modes trade precision for recall (RQ2, Sections 4.2 and 4.3), and how the training-free classifier and the cost of the funnel compare with the alternatives (RQ3, Sections 4.4 and 4.5). Two findings about the evaluation data follow (Sections 4.6 and 4.7). Unless stated otherwise, the funnel runs in high-precision mode, and results on the gold corpus use its provisional LLM labels.

4.1 Contribution of each stage (RQ1)

Each stage does a different job (Table 3 and Figure 3). Stage A removes many sentences only where sections are informative, Stage B raises precision about threefold while keeping about 90% of the positives, and Stage C provides most of the gain in precision at some cost in recall.

- **Stage A.** On SAL-40, keeping only Discussion and Conclusion sections removes 76% of the sentences and loses 2 of the 205 limitations. On the gold corpus, where Stage A is only a junk filter, it removes 18% of the sentences and 3% of the gaps.
- **Stage B.** The NLI classifier raises precision by a factor of 2.7 to 3.6 and keeps 88–91% of all positives, close to the 90% recall that its threshold rule targets on development data.
- **Stage C.** The LLM filter raises precision by a further 37 points on SAL-40, 28 on FWS and 63 on the gold corpus, and lowers recall by 6, 12 and 34 points, respectively. On the gold corpus, the lost recall falls mostly on research gaps and limitations (Section 4.3).

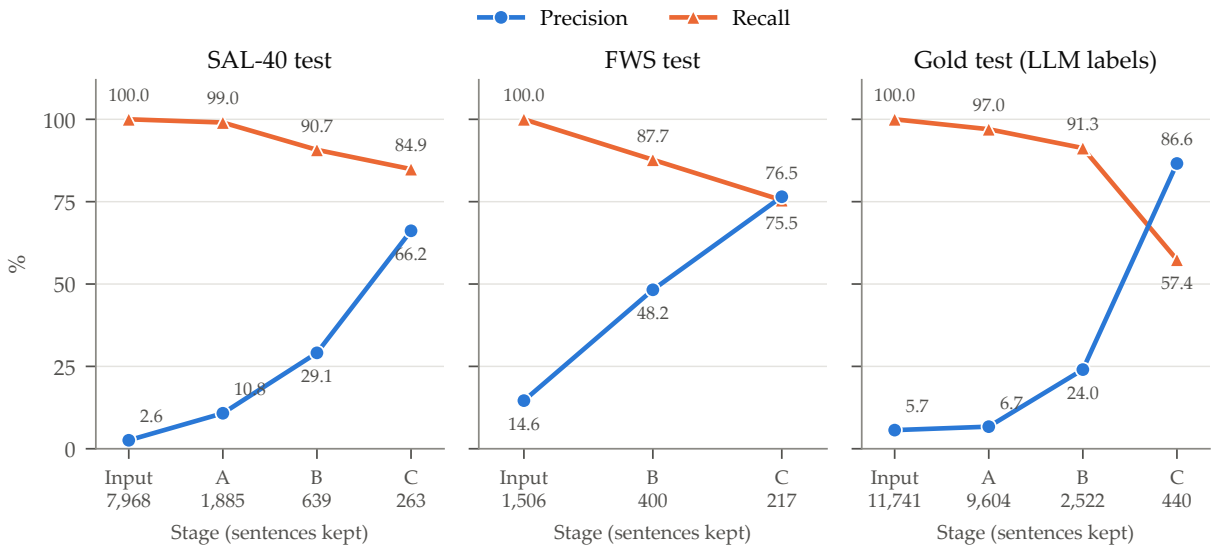


Figure 3: Precision and recall after each stage in high-precision mode. The number below each stage is the number of sentences it keeps. Gold-corpus results use LLM labels; FWS has no Stage A.

Table 3: Sentences and positives kept after each stage, high-precision mode. Precision and recall in %. Gold results use LLM labels.

Dataset	After stage	Sentences	Positives	Precision	Recall
SAL-40 test	Input	7968	205	2.6	100
	A. Discussion and Conclusion sections	1885	203	10.8	99.0
	B. NLI, $\tau = 0.7286$	639	186	29.1	90.7
	C. LLM filter	263	174	66.2	84.9
FWS test	Input (no Stage A)	1506	220	14.6	100
	B. NLI, $\tau = 0.1922$	400	193	48.2	87.7
	C. LLM filter	217	166	76.5	75.5
Gold test	Input	11,741	664	5.7	100
	A. Junk filter	9604	644	6.7	97.0
	B. NLI, $\tau = 0.354$	2522	606	24.0	91.3
	C. LLM filter	440	381	86.6	57.4

4.2 Precision control with three filter modes (RQ2)

Changing the final instruction of the prompt moves the filter along the precision–recall trade-off in the intended direction on every evaluation set (Table 4 and Figure 4): from high recall to balanced to high precision, precision rises and recall falls. In high-precision mode, precision is 66.2% on SAL-40, 76.5% on FWS and 86.6% on the gold corpus (83.4% on its 28 arXiv papers and 97.1% on its six biomedical papers).

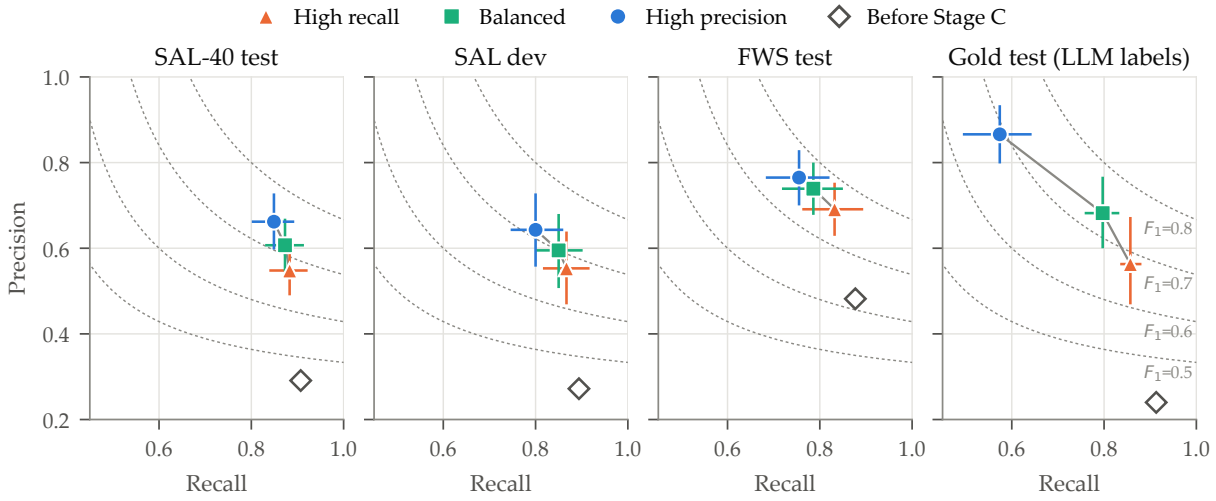


Figure 4: Precision against recall for the three filter modes and for the Stage B output before filtering, with 95% confidence intervals from resampling papers. Dotted curves connect points of equal F_1. Gold results use LLM labels.

Which mode has the highest F_1 depends on the dataset. On SAL-40, high precision is best (F_1 0.744, against 0.716 for balanced and 0.677 for high recall). On FWS, the three modes are within 0.007 of each other. On the gold corpus, balanced is best: its F_1 is higher than that of high recall by 0.055 (paired 95% CI 0.025 to 0.080) and higher than that of high precision by 0.045 (0.006 to 0.095), whereas high precision and high recall do not differ significantly (+0.011, −0.054 to +0.060).

Two checks indicate that this ordering of the modes is not specific to the test data. First, on SAL dev, whose papers influenced no prompt, the modes behave as on SAL-40 with slightly lower values: high-precision mode reaches 64.3% precision, 80.0% recall and F_1 0.713. Second, on the gold validation split, where the modes were checked before the balanced and high-precision test runs, high-precision mode kept 60.3% of the gaps it received and removed 97.7% of the non-gaps, compared with 94.1% and 82.9% in high-recall mode; precision was 90.4% and 66.8%, respectively.

Robustness. Two further checks concern the gold test split. Removing the 196 candidates used during development (Section 3.4) lowers F_1 slightly in every mode: from 0.690 to 0.675 in high-precision mode, from 0.735 to 0.723 in balanced mode and from 0.679 to 0.664 in high-recall mode. Repeating the high-recall run with the same inputs and seed changed 32 of the 2522 verdicts (1.3%), five of them on gaps; F_1 was 0.678 in one run and 0.679 in the other.

4.3 Effect of the modes by gap type

On the gold corpus, the recall that high-precision mode gives up is lost mainly on research gaps and limitations (Figure 5 and Table 5). Future-work recall barely changes across modes (82.5–86.6%), whereas research-gap recall falls from 89.3% in high-recall mode to 51.3% in high-precision mode, and limitation recall falls from 79.5% to 55.9%. This matters for the pooled

Table 4: Results of the three filter modes. Precision and recall in %, with 95% confidence intervals from resampling papers. Gold results use LLM labels. Kept: number of sentences returned.

Evaluation set	Setting	Kept	Precision	Recall	F_1
SAL-40 test (human labels)	Before Stage C	639	29.1 (26.2–32.1)	90.7 (86.7–94.5)	0.441 (0.404–0.475)
	High precision	263	66.2 (59.7–72.8)	84.9 (80.1–89.3)	0.744 (0.695–0.789)
	Balanced	295	60.7 (54.6–66.9)	87.3 (83.0–91.4)	0.716 (0.670–0.760)
	High recall	330	54.8 (49.0–61.3)	88.3 (83.9–92.2)	0.677 (0.627–0.725)
SAL dev (human labels)	Before Stage C	591	27.2 (23.3–31.5)	89.4 (85.1–93.5)	0.418 (0.369–0.468)
	High precision	224	64.3 (55.7–72.8)	80.0 (74.6–86.0)	0.713 (0.647–0.774)
	Balanced	257	59.5 (50.7–68.0)	85.0 (79.9–90.2)	0.700 (0.628–0.764)
	High recall	282	55.3 (46.9–63.9)	86.7 (81.6–91.7)	0.675 (0.602–0.742)
FWS test (human labels)	Before Stage C	400	48.2 (42.2–54.5)	87.7 (81.1–93.4)	0.623 (0.565–0.677)
	High precision	217	76.5 (70.0–82.9)	75.5 (68.3–82.1)	0.760 (0.705–0.807)
	Balanced	234	73.9 (67.8–80.0)	78.6 (71.8–85.0)	0.762 (0.709–0.808)
	High recall	265	69.1 (62.9–75.3)	83.2 (76.2–89.4)	0.755 (0.702–0.804)
Gold test, all (LLM labels)	High precision	440	86.6 (79.8–93.4)	57.4 (49.4–64.3)	0.690 (0.619–0.748)
	Balanced	776	68.2 (60.0–76.7)	79.7 (75.8–83.3)	0.735 (0.678–0.789)
	High recall	1,011	56.3 (46.9–67.3)	85.7 (83.5–88.1)	0.679 (0.604–0.756)
Gold test, arXiv (LLM labels)	High precision	337	83.4 (76.0–91.3)	53.3 (44.4–61.1)	0.650 (0.573–0.713)
	Balanced	643	64.2 (56.0–73.6)	78.4 (73.8–82.6)	0.706 (0.647–0.765)
	High recall	862	51.7 (42.8–62.7)	84.6 (82.1–87.2)	0.642 (0.567–0.719)
Gold test, biomedical (LLM labels)	High precision	103	97.1 (94.5–99.0)	73.0 (63.7–81.0)	0.833 (0.761–0.885)
	Balanced	133	87.2 (82.4–92.2)	84.7 (80.5–87.7)	0.859 (0.833–0.889)
	High recall	149	82.6 (80.2–86.1)	89.8 (85.7–94.4)	0.860 (0.839–0.894)

scores: research gaps make up about half of the gold labels and are the type the earlier stages retain best, so they dominate the scores that pool all types. When research-gap sentences are excluded from scoring, high-precision mode has the highest F_1 of the three modes (0.701, against 0.595 for balanced and 0.509 for high recall).

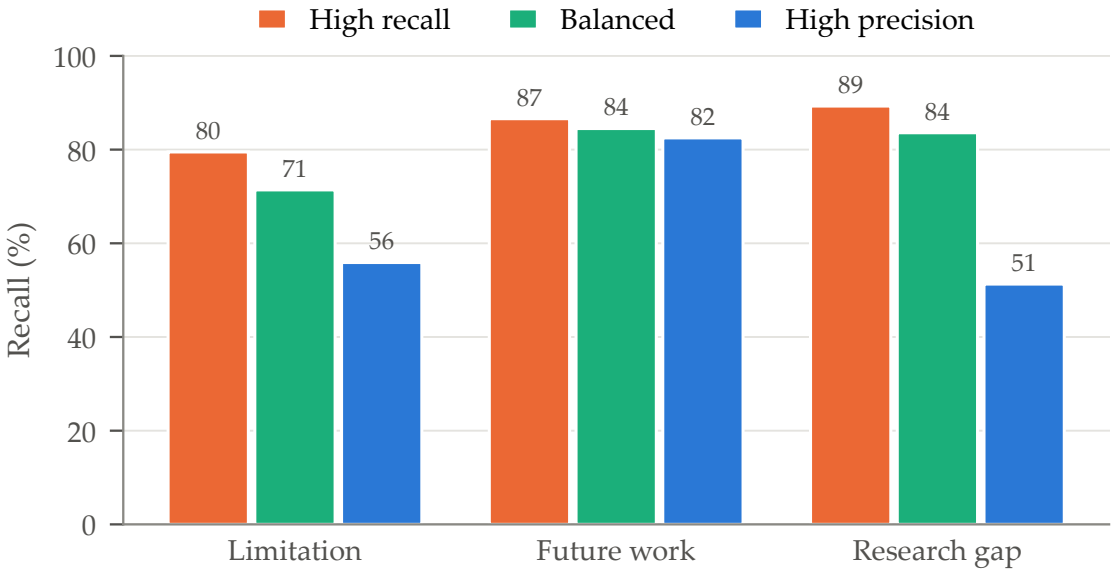


Figure 5: Recall by gap type on the gold test split (LLM labels) in each filter mode, in %.

Table 5: Gold test split (LLM labels) by gap type: recall for each type, and scores when research-gap sentences are excluded from scoring. In % except F_1.

	Recall by type			Without research gaps		
Mode	**Limitation (220)**	**Future work (97)**	**Research gap (347)**	**Precision**	**Recall**	F_1
High precision	55.9	82.5	51.3	77.5	64.0	0.701
Balanced	71.4	84.5	83.6	49.2	75.4	0.595
High recall	79.5	86.6	89.3	36.9	81.7	0.509

4.4 Training-free versus supervised classification (RQ3)

The training-free NLI classifier is a competitive Stage B. On SAL-40, the funnel built on it reaches almost the same F_1 as one built on a supervised PubMedBERT trained on SAL's own labels, although the supervised version is more precise. On the gold corpus, it performs better than or comparably to the alternatives we tested.

Against a supervised classifier on SAL. We ran the funnel in all three modes with PubMedBERT as Stage B. Before Stage C, PubMedBERT selects 391 candidates with 46.0% precision and 87.8% recall (F_1 0.604); it was the best of the eight SAL-trained classifiers (SciBERT 0.553, DistilBERT 0.523, frozen encoders 0.371–0.457). After Stage C in high-precision mode, it reaches 74.5% precision (95% CI 68.4–80.7), 78.5% recall and F_1 0.765. Table 6 compares the two classifiers mode by mode. With PubMedBERT, the funnel is 6.9 to 9.0 points more precise in every mode. In high-precision mode, however, the F_1 difference is small and not significant (−0.021, CI −0.053 to +0.014); in the other two modes, PubMedBERT's F_1 advantage is larger (0.035 and 0.059). The NLI classifier uses no SAL labels at all: its only SAL-specific setting is the threshold τ.

Table 6: SAL-40 test. Left: difference between the funnel with NLI and with PubMedBERT as Stage B (NLI minus PubMedBERT) in the same mode, paired over resampled papers; precision differences in points, 95% CI in brackets. Right: precision of the NLI funnel's output when scored against the LLM labels instead of SAL's human labels (Section 4.6), in %.

	NLI minus PubMedBERT		Precision against LLM labels	
Mode	**Precision**	F_1	**Limitation**	**Any gap type**
High precision	−8.4 (−11.8, −4.5)	−0.021 (−0.053, +0.014)	93.2	95.1
Balanced	−6.9 (−11.5, −2.3)	−0.035 (−0.068, 0.000)	88.8	91.2
High recall	−9.0 (−12.7, −5.1)	−0.059 (−0.089, −0.027)	81.8	85.2

Against other options on the gold corpus. On the half of the gold validation papers that was not used to choose the hypotheses, the three short hypotheses reached 39.2% precision at about 79% recall, whereas the bge-small classifier trained on the gold training split reached 28.2% precision at the same recall. On the test split, the funnel in high-precision mode has a higher F_1 than the ModernBERT model used alone with a single long hypothesis (0.690 against 0.614; difference 0.076, CI 0.020 to 0.132). It does not differ significantly from a funnel that uses the bge-small classifier as Stage B with the high-recall filter (F_1 0.665; difference +0.025, CI −0.037 to +0.087).

4.5 Cost (RQ3)

The funnel sends far fewer tokens to the LLM than judging every sentence, but only moderately fewer than a single whole-paper call (Table 7 and Figure 6). Compared with judging every sentence, it sends 3.7 (FWS) to 13.1 (SAL-40) times fewer input tokens. Compared with one whole-paper call, the saving is 3.1 times on SAL-40, 2.0 times on FWS and 1.25 times on the gold corpus. The saving is small on arXiv papers for two reasons: Stage B passes about one sentence in five, and each candidate is sent with its neighbouring sentences, so the funnel sends almost as much text as the whole paper. Whether the funnel is still preferable there depends on whether whole-paper prompting finds the same gaps, which the comparison in progress will show. The token counts of the other two modes differ from those of high-precision mode by less than 2%, because only the final instruction changes.

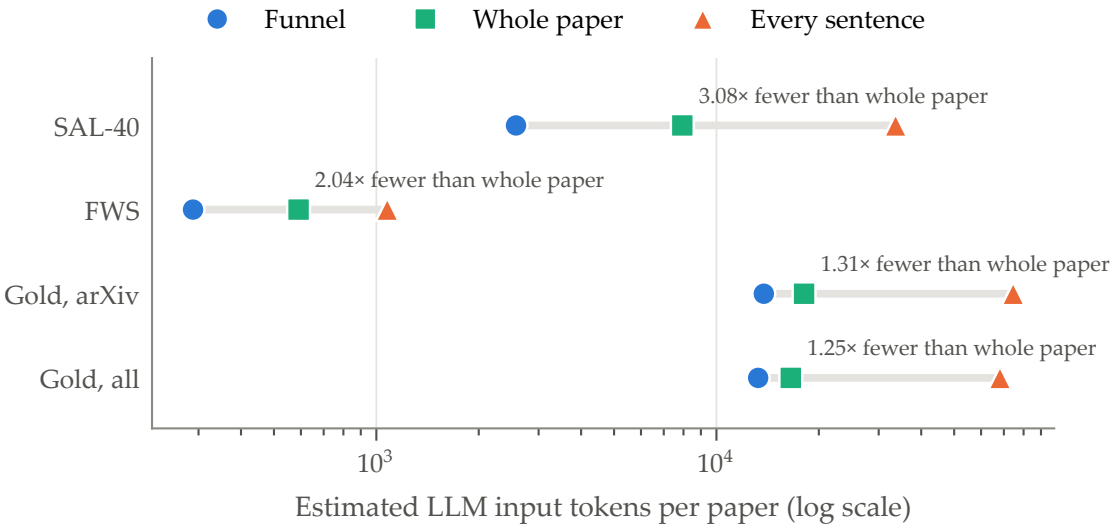


Figure 6: Estimated LLM input tokens per paper (`cl100k_base`, log scale) for the funnel in high-precision mode, one whole-paper call, and judging every sentence in batches.

Table 7: Estimated LLM input tokens (`cl100k_base`), high-precision mode, all test papers. Calls: number of LLM requests made by the funnel. Saving: the alternative's tokens divided by the funnel's tokens.

Dataset, Stage B	Candidates	Calls	Funnel	Whole paper	Every sentence	Saving vs whole paper	Saving vs every sentence
SAL-40, NLI	639	32	102,978	317,475	1,345,409	3.08	13.07
SAL-40, PubMedBERT	391	20	62,835	317,475	1,345,409	5.05	21.41
FWS, NLI	400	20	60,410	123,445	225,015	2.04	3.72
Gold test, NLI	2,522	127	450,955	562,440	2,318,416	1.25	5.14
Gold test (arXiv), NLI	2,137	107	385,897	506,753	2,087,907	1.31	5.41

4.6 Precision depends on the label source

The precision of the same output changes greatly with the reference labels. The right side of Table 6 scores the SAL-40 output against SAL's human labels and against our LLM labels. Of the 263 sentences returned in high-precision mode, SAL's annotators marked 174 as limitations, whereas the LLM ensemble marked 245 as limitations and 250 as a gap of some type. The two label sets disagree mainly on 77 sentences that are gaps according to the LLM labels but not according to SAL; only 12 returned sentences are negative under both. Precision is therefore 66.2% against SAL and 93.2–95.1% against the LLM labels. The true precision depends on which label set is right for these 77 sentences, and the human study described in Section 5.2 is designed to settle this.

4.7 Letter case reveals the label in FWS

In the FWS recognition corpus as we obtained it (64,896 sentences from 9013 papers), all 9009 positive sentences are entirely lowercase, and almost no negative sentence is. The trivial rule "a sentence is positive if it is all lowercase" therefore reaches 99.8% precision and 100% recall. Our copy has the same size in bytes as the published file [29], so the artefact appears to be present in the released corpus, although we did not compare the two files directly. Any case-sensitive model, and any LLM that sees the raw text, can exploit this signal. All FWS results above therefore use lowercased text, and we do not use an earlier FWS result of ours that was obtained on raw text.

5 Discussion

We first answer the research questions, then discuss how reliable the extracted gaps are, why section-based localisation does not transfer to computer-science papers, and what our results imply for evaluation data. The section ends with the limitations of this study.

5.1 Answers to the research questions

RQ1: stage contribution. The three stages have distinct roles. Stage A is cheap: when section headings are informative, as in RCT reports, it removes three quarters of the text at almost no cost in recall. Stage B largely determines the final recall, because Stage C only removes candidates and cannot recover a gap that Stage B missed; with our threshold rule, this ceiling was 88–91%. Stage C provides most of the gain in precision, between 28 and 63 points. Throughout, every reported gap remains a sentence the authors wrote.

RQ2: precision control. Changing a single instruction moves the filter along the precision–recall trade-off in the expected order on every evaluation set: SAL-40, SAL dev, FWS, and the

gold test split as a whole and in its arXiv and biomedical parts. The same order appeared on the gold validation split before the balanced and high-precision modes were run on test papers. The funnel can therefore be adapted to its use. High precision suits idea generation, where every false gap costs a downstream model or a person time; high recall suits a reviewer who wants to see every candidate. On human labels, high precision costs no F_1: it is the best mode on SAL-40 and level with the others on FWS. On the LLM-labelled corpus, it does lose recall, mostly on research gaps. A plausible explanation is that the high-precision instruction asks the model to drop background and motivation (Appendix A), and authors often state open problems in exactly that role, when they motivate their own work.

RQ3: classifier and cost. A zero-shot NLI classifier with short hypotheses that name the gap type is a workable Stage B. It needs no gap-labelled training data, works on both biomedical and computer-science text, and on SAL comes within 0.021 F_1 of a PubMedBERT trained on SAL's own labels. The supervised classifier is, however, 8.4 points more precise in high-precision mode, so a project that has in-domain labels should still use them. On cost, the funnel sends 3.7 to 13 times fewer tokens than judging every sentence, but its saving over a single whole-paper call is modest and almost disappears on arXiv papers. A whole-paper prompt could also include the mode instruction and return sentence numbers. On arXiv papers, the funnel is therefore worthwhile only if it extracts gaps more accurately than whole-paper prompting; the comparison in progress will show whether it does.

5.2 How reliable are the extracted gaps?

How good the extracted gaps are depends on which labels are trusted. On SAL-40, high-precision mode reaches 66.2% precision against SAL's labels and 93.2% against the LLM labels (Section 4.6). Either label set may be wrong on the 77 sentences where they disagree. SAL's annotators may have left some limitation statements unlabelled; the LLM labels, in turn, are inclusive by construction, because adjudication added sentences that only one model had marked. The true precision is therefore likely to lie between the two values.

Label errors may also affect the gold corpus. In an earlier high-recall run during development, which is not part of the results above, five of the 34 test papers accounted for 59% of the false positives. Each of these papers had only about 18–24 gold gaps despite having 220 to 824 sentences, which may indicate that long papers are under-labelled.

To measure precision against human judgement, we have designed a blind human study. Two raters and an independent tie-breaker will label a frozen sample of 135 sentences, drawn from the high-precision output on arXiv and SAL-40 and from decoy sentences. Its results will be added in a later version.

5.3 Section-based localisation does not transfer to computer-science papers

In RCT reports, limitations cluster in the Discussion and Conclusion sections, so a section rule works well. In arXiv papers it did not, in our development work. An earlier localiser kept only the sections where gaps are usually stated, such as the discussion and conclusion. It retained only 55.6% of the arXiv gold gaps when sections were inferred from the extracted text, and 55.2% when sections were taken from the LaTeX sources of 175 papers. More accurate section information therefore did not help: with LaTeX sections, 82% of the missed gaps were in methods, experiments or results sections. These are development observations from a single run that we have not recomputed for this version. For arXiv papers, we therefore use only the junk filter as Stage A, which keeps 97% of the gaps, and leave the narrowing to Stage B.

5.4 Evaluation data needs checking

The FWS case artefact (Section 4.7) shows that a published benchmark can contain a trivial signal that a modern model will find. FWS results from case-sensitive models, or from LLMs given the raw text, including an earlier result of ours, should be treated with caution unless the text is normalised. More generally, the precision we measure on SAL-40 changes more with the label source than with the choice of Stage B classifier (Section 4.6). Stating which labels a number rests on is therefore as important as reporting the number.

5.5 Limitations

- **LLM labels.** All results on the gold corpus rest on labels produced by Claude models. Human validation has been designed and its sample frozen, but it has not yet been carried out. The agreement between the two labelling models ($\kappa = 0.778$) measures their consistency, not their accuracy.
- **Development on test papers.** The Stage C model, batch size, use of context and high-recall instruction were chosen on 196 gold test candidates, and the balanced and high-precision instructions were written after reading false positives on gold test papers. We report results without the 196 candidates, ran and report all modes on every dataset, and checked the modes on the validation split first; still, the gold test split is not untouched. SAL dev provides the cleanest held-out check, and its results agree with those on SAL-40.
- **Scope of the benchmarks.** SAL covers RCT reports only, FWS covers NLP conclusion sections only, and each labels a single gap type. The six biomedical papers in the gold test split are also SAL papers.
- **Research gaps.** Half of the gold labels are research gaps, which the earlier stages retain better than the other types; we therefore also report recall by type and scores without research gaps.
- **Comparisons in progress.** Whole-paper LLM prompting on the same papers and reproductions of published methods are not yet included.
- **Cost.** Token counts are estimates with a vendor-neutral tokenizer and cover LLM input only.
- **One LLM.** Stage C uses a single model; other models may give different results. Repeating one run changed 1.3% of the verdicts.

6 Conclusions

We presented a three-stage funnel that extracts author-stated research gaps as unmodified source sentences: rules remove text that is unlikely to contain gaps, a zero-shot NLI classifier selects candidates without gap-labelled training data, and an LLM filter removes false positives in one of three modes set by a single instruction.

On the human-labelled benchmarks, high-precision mode reached 66.2% precision at 84.9% recall for limitations in RCT reports (SAL-40) and 76.5% precision at 75.5% recall for future work in NLP conclusion sections (FWS), and held-out SAL papers gave similar results. On a 230-paper corpus with provisional LLM labels for three gap types, it reached 86.6% precision at 57.4% recall. The modes shifted precision and recall in the intended direction on every dataset. The training-free classifier came close to a supervised one on SAL in F_1, though not in precision. The funnel is clearly cheaper than judging every sentence, but only moderately cheaper than a single whole-paper call.

The next steps are the human validation of the LLM-labelled corpus, comparisons with whole-paper prompting and published methods, and the use of the extracted gaps for automated research-idea generation.

Data Availability

Code, prompts, evaluation scripts, the gold-corpus labels and paper splits, and the exact SAL and FWS evaluation splits are openly available at `https://github.com/Yazangthb/Gap2Idea` [41]. The gold-corpus labels were produced by language models and have not yet been validated by people. The SAL data [28] and the FWS corpus [29] were created by their authors and remain subject to their original terms; copies are included in the repository so that the evaluation can be reproduced. The arXiv papers are available from arXiv; the repository includes the text we extracted from them.

Use of Generative AI

During the preparation of this study and manuscript, the authors used Claude Opus 5.5 (Anthropic), through Claude Code, to write and run analysis code and to draft and edit text. The authors have reviewed and edited the output and take full responsibility for the content of this publication.

Funding and Conflicts of Interest

This research received no external funding. The authors declare no conflicts of interest.

A Stage C prompt

The system prompt used on the gold corpus is reproduced below. The three modes differ only in the paragraph marked [MODE], whose three versions are listed after the prompt. For SAL and FWS, the decision rule keeps only item (a) or item (b), respectively; the other two items are added to the list of sentences to drop, and the words "gap" and "research gap" are replaced by "limitation" or "future-work direction". All prompts are in the code repository (`scripts/bench/stage_c_rubric_val.py`, `stage_c_sal40_modes.py`, `nli_sal_fws.py`).

> You are a precision filter for a research-gap extraction pipeline. Judge each candidate sentence: does it genuinely state a research gap?
>
> Each item has: "i" (integer id), "type" (the predicted gap type), "sentence" (the candidate to judge), and "context" (surrounding text). Judge the "sentence", using "context" only to disambiguate.
>
> Decision — answer YES (keep) if the sentence states ANY of: (a) a limitation / weakness / caveat / bias / constraint / assumption of the authors' OWN work (e.g., small sample, no blinding, single setting, short follow-up, confounding, limited generalizability, unvalidated measure, "we did not / could not . . . ", "a limitation is . . . "); (b) an explicit FUTURE-WORK direction the authors propose ("in future work we will . . . ", "future research should . . . ", "it would be interesting to . . . "); (c) an unsolved problem / missing capability / open research gap in the field or prior work ("it remains unknown . . . ", "no existing method . . . ", "current approaches struggle with . . . ").
>
> Answer NO (drop) if the sentence is instead: a result or finding, a method/experimental-setup description, a positive or contribution claim ("we propose/show/achieve . . . "), background or related-work description, a citation or reference

fragment, acknowledgements / funding / author contributions / competing interests, a figure or table caption, a dataset/metric definition, or a generic non-gap statement.

[MODE]

Return JSON of the form {"results":[{"id":<i>, "keep":<bool>}, . . .]} with EXACTLY one entry per input item, where id is the item's "i" and keep=true means YES.

The [MODE] paragraph:

- **High recall:** "Be KEEP-BIASED: when you are genuinely uncertain whether a sentence is a real gap, answer YES. Only answer NO when it clearly is NOT a gap by the rules above."
- **Balanced:** "Judge each sentence on the rules above, without favouring YES or NO."
- **High precision:** "Be STRICT. Answer YES only when the sentence itself explicitly states the limitation, the proposed direction, or the unsolved problem. Also answer NO for: a general property or difficulty of a method or domain given as background or motivation; a statement of why the authors' approach is needed; an observed experimental outcome; and a passing remark that names no specific gap. When genuinely uncertain, answer NO."

References

[1] Kilicoglu, H.; Rosemblat, G.; Malički, M.; ter Riet, G. Automatic Recognition of Self-Acknowledged Limitations in Clinical Research Literature. *Journal of the American Medical Informatics Association* **2018**, *25*, 855–861. https://doi.org/10.1093/jamia/ocy038.

[2] Zhu, Z.; Wang, D.; Shen, S. Recognizing Sentences Concerning Future Research from the Full Text of JASIST. *Proceedings of the Association for Information Science and Technology* **2019**, *56*, 858–859. https://doi.org/10.1002/pra2.206.

[3] Zhang, C.; Xiang, Y.; Hao, W.; Li, Z.; Qian, Y.; Wang, Y. Automatic Recognition and Classification of Future Work Sentences from Academic Articles in a Specific Domain. *arXiv preprint arXiv:2212.13860* **2022**. Preprint version used for source verification.

[4] Xu, Z.; Zhao, Y.; Patwardhan, M.; Vig, L.; Cohan, A. Can LLMs Identify Critical Limitations within Scientific Research? A Systematic Evaluation on AI Research Papers. In Proceedings of the Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers). Association for Computational Linguistics, 2025, pp. 20652–20706. https://doi.org/10.18653/v1/2025.acl-long.1009.

[5] Mishra, S.; Rajeev, N.; Chakraborty, T. Tree-of-Concerns: Hierarchical Multi-Agent Debate for Unstated-Limitation Extraction in Scientific Critique. *arXiv preprint arXiv:2608.20777* **2026**. Accepted to Findings of EMNLP 2026.

[6] Lan, M.; Cheng, M.; Hoang, L.; ter Riet, G.; Kilicoglu, H. Automatic Categorization of Self-Acknowledged Limitations in Randomized Controlled Trial Publications. *Journal of Biomedical Informatics* **2024**, *152*, 104628. https://doi.org/10.1016/j.jbi.2024.104628.

[7] Teufel, S.; Moens, M. Summarizing Scientific Articles: Experiments with Relevance and Rhetorical Status. *Computational Linguistics* **2002**, *28*, 409–445. https://doi.org/10.1162/089120102762671936.

[8] Liakata, M.; Teufel, S.; Siddharthan, A.; Batchelor, C. Corpora for the Conceptualisation and Zoning of Scientific Papers. In Proceedings of the Proceedings of the Seventh International Conference on Language Resources and Evaluation (LREC'10), Valletta, Malta, 2010.

[9] Fisas, B.; Ronzano, F.; Saggion, H. A Multi-Layered Annotated Corpus of Scientific Papers. In Proceedings of the Proceedings of the Tenth International Conference on Language Resources and Evaluation (LREC'16), Portorož, Slovenia, 2016; pp. 3081–3088.

[10] Dernoncourt, F.; Lee, J.Y. PubMed 200k RCT: A Dataset for Sequential Sentence Classification in Medical Abstracts. In Proceedings of the Proceedings of the Eighth International Joint Conference on Natural Language Processing (Volume 2: Short Papers), Taipei, Taiwan, 2017; pp. 308–313.

[11] Cohan, A.; Beltagy, I.; King, D.; Dalvi, B.; Weld, D. Pretrained Language Models for Sequential Sentence Classification. In Proceedings of the Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP), Hong Kong, China, 2019; pp. 3693–3699. https://doi.org/10.18653/v1/D19-1383.

[12] Dayrell, C.; Candido Jr., A.; Lima, G.; Machado Jr., D.; Copestake, A.; Feltrim, V.; Tagnin, S.; Aluisio, S. Rhetorical Move Detection in English Abstracts: Multi-label Sentence Classifiers and their Annotated Corpora. In Proceedings of the Proceedings of the Eighth International Conference on Language Resources and Evaluation (LREC'12), Istanbul, Turkey, 2012; pp. 1604–1609.

[13] Stead, C.; Smith, S.; Busch, P.; Vatanasakdakul, S. Emerald 110k: A Multidisciplinary Dataset for Abstract Sentence Classification. In Proceedings of the Proceedings of the 17th Annual Workshop of the Australasian Language Technology Association, Sydney, Australia, 2019; pp. 120–125.

[14] Bolaños, F.; Salatino, A.; Osborne, F.; Motta, E. Modelling and Classifying the Components of a Literature Review. *arXiv preprint arXiv:2508.04337* **2025**.

[15] Jin, D.; Szolovits, P. Hierarchical Neural Networks for Sequential Sentence Classification in Medical Scientific Abstracts. In Proceedings of the Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP), Brussels, Belgium, 2018; pp. 3100–3109. https://doi.org/10.18653/v1/D18-1349.

[16] Brack, A.; Entrup, E.; Stamatakis, M.; Buschermöhle, P.; Hoppe, A.; Ewerth, R. Sequential Sentence Classification in Research Papers Using Cross-Domain Multi-Task Learning. *International Journal on Digital Libraries* **2024**, *25*, 377–400. https://doi.org/10.1007/s00799-023-00392-z.

[17] Lan, M.; Zheng, L.; Ming, S.; Kilicoglu, H. Multi-label Sequential Sentence Classification via Large Language Model. In Proceedings of the Findings of the Association for Computational Linguistics: EMNLP 2024. Association for Computational Linguistics, 2024, pp. 16086–16104. https://doi.org/10.18653/v1/2024.findings-emnlp.944.

[18] Zhang, Y.; Zhang, C. Extracting Problem and Method Sentence from Scientific Papers: A Context-Enhanced Transformer Using Formulaic Expression Desensitization. *Scientometrics* **2024**, *129*, 3433–3468. https://doi.org/10.1007/s11192-024-05048-6.

[19] Hao, W.; Li, Z.; Qian, Y.; Wang, Y.; Zhang, C. The ACL FWS-RC: A Dataset for Recognition and Classification of Sentence about Future Works. In Proceedings of the Proceedings of the ACM/IEEE Joint Conference on Digital Libraries (JCDL), 2020, pp. 261–269. https://doi.org/10.1145/3383583.3398526.

[20] Suray, J.; Klemmer, J.H.; Schmüser, J.; Fahl, S. How the Future Works at SOUPS: Analyzing Future Work Statements and Their Impact on Usable Security and Privacy Research. *arXiv preprint arXiv:2405.20785* **2024**.

[21] Thaepprasit, T.; Kehasukcharoen, P.; Wang, D.; Denton, R.; Vateekul, P.; Lertvittayakumjorn, P. What Limits Us? Analyzing Self-Reported Limitations in NLP Research. *arXiv preprint arXiv:2609.15191* **2026**.

[22] Al Azher, I.; Mokarrama, M.J.; Guo, Z.; Ray Choudhury, S.; Alhoori, H. BAGELS: Benchmarking the Automated Generation and Extraction of Limitations from Scholarly Text. In Proceedings of the Findings of the Association for Computational Linguistics: EMNLP 2025. Association for Computational Linguistics, 2025, pp. 19279–19294. https://doi.org/10.18653/v1/2025.findings-emnlp.1050.

[23] Al Azher, I.; Mokarrama, M.J.; Guo, Z.; Ray Choudhury, S.; Alhoori, H. FutureGen: A RAG-based Approach to Generate the Future Work of a Scientific Article. *arXiv preprint arXiv:2503.16561* **2025**.

[24] Cheng, J.; Liu, X.; Tao, Z.; Gu, H.; Wang, Y. Research and Application of Future Work Sentence Extraction Methods Based on Large Language Models: Taking Papers in the Natural Language Processing (NLP) Field as Example. *Science Focus* **2026**, *21*, 81–98. In Chinese; English title and abstract supplied by the publisher, https://doi.org/10.15978/j.cnki.1673-5668.20250721.

[25] Salem, N.M.; White, E.; Bada, M.; Hunter, L. GAPMAP: Mapping Scientific Knowledge Gaps in Biomedical Literature Using Large Language Models. *arXiv preprint arXiv:2510.25055* **2025**.

[26] Faizullah, A.R.B.M.; Urlana, A.; Mishra, R. LimGen: Probing the LLMs for Generating Suggestive Limitations of Research Papers. In Proceedings of the Machine Learning and Knowledge Discovery in Databases. Research Track (ECML PKDD 2024). Springer, 2024, Lecture Notes in Computer Science, pp. 106–124. https://doi.org/10.1007/978-3-031-70344-7_7.

[27] Yin, W.; Hay, J.; Roth, D. Benchmarking Zero-shot Text Classification: Datasets, Evaluation and Entailment Approach. In Proceedings of the Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP). Association for Computational Linguistics, 2019, pp. 3914–3923. https://doi.org/10.18653/v1/D19-1404.

[28] Lan, M. SAL_Type_Classification: Data and Code for Self-Acknowledged Limitation Categorization. GitHub repository, 2024.

[29] Xiang, Y. FWS: Future Work Sentence Recognition and Classification Data. GitHub repository, recognition corpus file, 2022.

[30] Cohen, J. A Coefficient of Agreement for Nominal Scales. *Educational and Psychological Measurement* **1960**, *20*, 37–46. https://doi.org/10.1177/001316446002000104.

[31] Warner, B.; Chaffin, A.; Clavié, B.; Weller, O.; Hallström, O.; Taghadouini, S.; Gallagher, A.; Biswas, R.; Ladhak, F.; Aarsen, T.; et al. Smarter, Better, Faster, Longer: A Modern Bidirectional Encoder for Fast, Memory Efficient, and Long Context Finetuning and Inference. *arXiv preprint arXiv:2412.13663* **2024**.

[32] tasksource. tasksource/ModernBERT-large-nli. Hugging Face model, revision ca476cb923a8637073d4ceb0f19f7fc236e260d4, 2025.

[33] Kwon, W.; Li, Z.; Zhuang, S.; Sheng, Y.; Zheng, L.; Yu, C.H.; Gonzalez, J.E.; Zhang, H.; Stoica, I. Efficient Memory Management for Large Language Model Serving with PagedAttention. In Proceedings of the Proceedings of the 29th Symposium on Operating Systems Principles (SOSP), 2023, pp. 611–626. https://doi.org/10.1145/3600006.3613165.

[34] Gu, Y.; Tinn, R.; Cheng, H.; Lucas, M.; Usuyama, N.; Liu, X.; Naumann, T.; Gao, J.; Poon, H. Domain-Specific Language Model Pretraining for Biomedical Natural Language Processing. *ACM Transactions on Computing for Healthcare* **2021**. https://doi.org/10.1145/3458754.

[35] Beltagy, I.; Lo, K.; Cohan, A. SciBERT: A Pretrained Language Model for Scientific Text. In Proceedings of the Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing. Association for Computational Linguistics, 2019, pp. 3615–3620. https://doi.org/10.18653/v1/D19-1371.

[36] Sanh, V.; Debut, L.; Chaumond, J.; Wolf, T. DistilBERT, a Distilled Version of BERT: Smaller, Faster, Cheaper and Lighter. *arXiv preprint arXiv:1910.01108* **2019**.

[37] Reimers, N.; Gurevych, I. Sentence-BERT: Sentence Embeddings Using Siamese BERT-Networks. In Proceedings of the Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing. Association for Computational Linguistics, 2019, pp. 3982–3992. https://doi.org/10.18653/v1/D19-1410.

[38] Xiao, S.; Liu, Z.; Zhang, P.; Muennighoff, N. C-Pack: Packaged Resources To Advance General Chinese Embedding. *arXiv preprint arXiv:2309.07597* **2023**.

[39] Efron, B.; Tibshirani, R.J. *An Introduction to the Bootstrap*; Chapman and Hall/CRC: New York, 1993. https://doi.org/10.1201/9780429246593.

[40] OpenAI. tiktoken. Software, cl100k_base encoding, 2023.

[41] Alnakri, Y.; Orabi, O.; Aldaghstany, H.S. Gap2Idea: Code for Extracting Author-Stated Research Gaps. GitHub repository, 2026.